INCH - POUND

MIL-PRF-53136
5 August 1996

PERFORMANCE SPECIFICATION

GUN LAYING AND POSITIONING SYSTEM (GLPS)

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers an automated Gun Laying and Positioning System (GLPS).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documentation cited in other sections of this specification or recommended for additional information or as examples. While effort has been made to ensure completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

ICD-GPS-153 - GPS User Equipment Interface Control Document for the RS-232/RS-422 Interface of DoD Standard GPS UE Radio Receivers

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Beneficial comments, recommendations, additions, deletions, clarifications etc., and any other data which may improve this document should be sent to: U.S. ARMY TOPOGRAPHIC ENGINEERING CENTER, ATTN CETEC-PD-TE, 7701 TELEGRAPH ROAD, ALEXANDRIA, VA 22315-3864.

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2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation.

American National Standards Institute, ANSI Z136.1

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 System description. The Gun Laying and Positioning System, hereinafter referred to as "GLPS", shall consist of the major components in 3.1.1 and additional components in 3.1.2. The azimuth gyroscope, electronic digital optical instrument, and laser rangefinder shall be fully integrated with an embedded microprocessor and control and display system for computation and display of gun laying and positioning data. Optics shall be integrated to the maximum extent possible. The Precision Lightweight Global Positioning System Receiver, hereafter referred to as "PLGR", shall be provided by the Government (see 3.1.1.4). The assembled GLPS shall be battery powered, mounted on a tripod, and integrated both physically and electronically without the use of external wires or cables (except for the PLGR, see 3.1.1.4). When assembled, the GLPS shall be capable of being precisely centered over a survey control point.

3.1.1 Major component list and characteristics

3.1.1.1 Azimuth gyroscope. The azimuth gyroscope shall be capable of automatically determining grid azimuths with respect to true north. It shall be integrated with the electronic digital optical instrument, laser rangefinder, battery/battery pack, illuminating target rod, and microprocessing, data control and display system without the use of external wires or cables.

3.1.1.2 Electronic Digital Optical instrument. The optical instrument shall be an electronic digital horizontal and vertical angle measuring instrument. It shall be integrated with the azimuth gyroscope, laser rangefinder, battery/battery pack, illuminating target rod, and microprocessing, data control and display system without the use of external wires or cables. The reticle shall be capable of being illuminated with an adjustable intensity. The reticle pattern shall be a cross, similar to that used in other U.S. field artillery reticles.

3.1.1.3 Laser Rangefinder. The laser rangefinder shall be eye safe (see 3.8). It shall be integrated with the azimuth gyroscope, electronic digital optical instrument, battery/battery pack, illuminating target rod, and microprocessing, data control and display system without the use of external wires or cables. Laser and optical instrument optics shall be integrated to the maximum extent possible.

3.1.1.4 Precision Lightweight Global Positioning System Receiver (PLGR). The PLGR will be provided by the Government to military units using the GLPS.

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It shall be powered by its own internal battery and mounted separately on the tripod with a mounting bracket provided by the contractor. An RS-232/ RS-422 interface cable shall be provided by the contractor to connect the PLGR to the GLPS (see 3.4.1). The PLGR shall be capable of being detached from the tripod for use elsewhere without affecting the GLPS performance.

3.1.1.5 Tripod. The tripod shall be a collapsable telescopic leg type with provisions for mounting the assembled GLPS and the separate PLGR. It shall have an adjustable height range from 35 ±2 to 53 ±2 inches and be as lightweight as possible, consistent with the needed stability and overall weight requirements of the GLPS. Tripod feet shall be pointed to prevent slipping on sloped and muddy surfaces. A shoulder sling shall be provided for carrying the tripod.

3.1.1.6 Transport case(s). The transport case(s) shall be capable of stowing all components, parts, materials, supplies, tools, and other items that are required for the GLPS to perform as specified herein (with exception of the PLGR and tripod). Each case shall be provided with a pressure relief valve and equipped with a carrying handle(s) for a one or two person lift, as appropriate (see 3.9.d). Case(s) shall be seamless or welded seam construction and capable of being decontaminated using military standard decontaminating procedures and materials. The case(s) shall protect the GLPS from the environmental and transportability conditions described in 3.5 and 3.6. The use of military standard cases is preferred.

3.1.1.7 Battery/battery pack. A single rechargeable battery or battery pack shall be used to power the GLPS (except the PLGR, see 3.1.1.4). A DoD approved battery/battery pack with a National Stock Number (NSN) is preferred. An additional battery/battery pack shall be provided as a spare. Battery enclosures shall be constructed so that battery venting or leakage will not damage other GLPS components. Batteries shall be attached to the GLPS without the use of external wires or cables. Batteries shall be easily removable when being stored or exchanged for a newly charged battery/battery pack. To remove and replace the battery/battery pack shall not take longer than one minute.

3.1.1.8 Illuminating target rod. The illuminating target rod shall be adapted to the GLPS without the use of external wires or cables. The target rod shall use a standard light bulb with a NSN. The bulb shall be easily accessible for replacement.

3.1.2 Additional component list and characteristics

3.1.2.1 Battery charger. A battery charger shall be supplied with each GLPS for the recharging of its batteries/battery packs. Cables with appropriate clips and/or plugs shall be provided with the battery charger (see 3.1.2.5).

3.1.2.2 Power conditioner. If a separate power conditioner is needed for operation of the GLPS from the military vehicle battery, one shall be provided with each GLPS (see 3.4.6).

3.1.2.3 Protective cover. A protective cover shall be provided with each GLPS. The cover shall fit over the electrical components when set up for operation and protect them from wind driven rain, ice, sand, and dust when not being operated. The cover shall be constructed of a lightweight material capable of use in high and low temperatures (see 3.5.1).

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3.1.2.4 Winterization kit. If a separate winterization kit is needed for operation at low temperatures (see 3.5.1), one shall be provided when specified (see 6.2).

3.1.2.5 Electrical cables, receptacles, and fuses. Electrical cables required for the GLPS operation from a back-up power source (see 3.4.6), for the battery charger (see 3.1.2.1), for the winterization kit (when provided, see 3.1.2.4), and for interface between GLPS and the PLGR (see 3.1.1.4) shall be provided with the GLPS. Cables shall be sized correctly to carry the power, be insulated with material that will withstand cold temperature to -46° C without breaking or cracking, and be provided with appropriate connecting clamps and plugs. Cables to the vehicle battery shall be provided with the NATO plug adapter. Electrical receptacles on GLPS components (including cables) shall be provided with protective covers when not in use. The protective covers shall be captive retained. Fuses shall be standard with NSNs and easily accessible for replacement.

3.1.2.6 Tripod base. When specified (see 6.2), a tripod base shall be supplied to securely position the tripod on hard surfaces, such as floors, and prevent damage to the surface. The tripod base is for use in training environments for demonstration and instructional purposes.

3.1.2.7 Tools, materials, supplies, and spares. Tools, materials, spares, supplies, and other items that are required for assembly, disassembly, operation, and maintenance of the GLPS shall be provided with each GLPS. When fuze and bulb replacement is part of operator maintenance, at least five spares of each type and rating shall be supplied. Additional spares may be provided at the discretion of the contractor.

3.2 First article. When specified (see 6.2), sample GLPSs (see 6.3), including major and additional components (see 3.1.1 and 3.1.2), shall be subjected to first article inspection in accordance with 4.2.

3.3 Materials. The contractor shall select the materials, but the materials shall be capable of meeting all of the operational and environmental requirements specified herein. The materials specified in MIL-P-11268 are recommended, but are not mandatory. Recovered materials shall be used to the maximum extent possible (see 6.4.1). Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the GLPS. Radioactive materials shall not be used without prior approval of the contracting office or designated representative. (Potential sources of radiation include luminous dials and markings, surge arresters, and optical lenses.)

3.3.1 Material deterioration prevention and control. The GLPS shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the GLPS may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier, such as tape or corrosion inhibiting sealant, which shall be compatible with the GLPS operating temperatures (see 3.5.1). MIL-STD-889 may be used for classification of dissimilar metals and protection when coupled. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

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3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and sub-components, and shall make information available upon request to the contracting officer or designated representative (see 6.2).

3.4 System performance. The GLPS shall enable the operator to rapidly self-locate and determine grid azimuth and position (coordinates and altitude) of each artillery howitzer from one centrally located orienting station. The GLPS shall be capable of laying howitzers by deflection angles with respect to grid north upon manual insertion of any azimuth of fire.

3.4.1 Interface requirements. The GLPS shall be capable of accepting Universal Transverse Mercator (UTM) coordinates and both mean sea level and datum height to the nearest meter from the PLGR via a RS-232/RS-422 cable interface (see reference ICD-GPS-153). The GLPS shall accept data in all PLGR set up modes to establish its position. The GLPS shall extract the operational datum from the PLGR and input this data into its memory. The GLPS shall also accept this data when entered manually through the data control and display panel.

3.4.2 Required Accuracies

3.4.2.1 Horizontal position. The GLPS shall be capable of positioning howitzers to an accuracy of 17.5 meters Circular Error Probable (CEP) horizontal after self-location using coordinates provided by the PLGR, coordinates obtained from sighting on a visible known survey control point (back polar plot method), or coordinates manually input from a known position. The GLPS shall be capable of self-location to an accuracy of 10.0 meters CEP horizontal by each of the above methods.

3.4.2.2 Vertical position. The GLPS shall be capable of positioning howitzers to an accuracy of 10 meters Probable Error (PE) vertical after self-location using the altitude provided by the PLGR or obtained from a back polar plot, or manually input from a known position.

3.4.2.3 Azimuth. The GLPS shall be capable of providing grid azimuths and calculating deflections from the orienting station to howitzers accurate to 0.4 mil (PE) between 0° and 65° North latitude and 0° and 65° South latitude; and accurate to 0.67 mil (PE) between 65° and 75° North and South latitudes.

3.4.2.4 Range. The laser rangefinder shall determine ranges between 30 meters and 2000 meters with an accuracy of ±3 meters without the use of reflectors or prisms.

3.4.3 Setup time. The GLPS shall not require pre-alignment. It shall be capable of being set up for operation in 5 minutes or less, and in 7 minutes when in mission-oriented protective posture (MOPP) IV (Setup is defined in 6.4.2). When a means to level the GLPS is provided, the time required for leveling shall be included in the setup time. When a winterization kit is used at low temperatures (see 3.1.2.4), the installation and warmup period shall not be included in the setup time.

3.4.4 Mission times. The GLPS shall be capable of deriving position and deflection to the first howitzer within 4 minutes after setup and power is applied to the gyroscope, and shall require not more than 1 additional minute to derive position and deflection for each subsequent howitzer. When a winter-

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ization kit is used at low temperatures (see 3.1.2.4), the warmup period shall not be included in the mission time.

3.4.5 Microprocessing, data control and display. The GLPS shall collect, compute, and display all required data (i.e., slope distance, vertical angle, grid azimuth, deflection angle, GLPS coordinates and elevation, and howitzer coordinates and elevations). The microprocessor shall have a non-volatile memory of adequate capacity to store all GLPS constants and operating parameters including at least eight data blocks for reference points, howitzer positions, position of the orientating station and other targets; and random access memory with enough capacity to perform required computations. The microprocessor shall provide initiation and control of a system self-test. The self-test shall display possible errors in operation and failures of components including but not limited to power source, gyroscope, rangefinder and optical instrument; and shall include a battery low charge warning to alert the operator when the battery/battery pack needs to be replaced. The low charge warning shall be displayed before the battery fails and the operator shall be able to complete a mission (see 6.4.3) before replacing the battery. The GLPS shall not lose data while any of the power sources are being correctly removed or replaced. If a memory battery is used, it must be rechargeable, or easily accessible for replacement with a NSN battery. Visual displays shall be as simple as possible and they shall be designed to:

a. provide only task essential information to operators and maintainers;

b. cue appropriate actions;

c. preclude ambiguities;

d. provide logical-intuitive steps to task accomplishment;

e. communicate directly; and

f. make comprehension easy.

The data control and display system shall conform to good human factors design (see 3.9.f). When specified (see 6.2), the contractor shall propose menu driven prompts and visual displays to the Government for approval prior to putting them into effect.

3.4.6 System Power. The battery/battery pack used to power the GLPS shall have sufficient ampere-hour capacity for the GLPS to conduct not less than 20 missions (see 6.4.3). The GLPS shall be capable of being operated using power supplied directly from a military vehicle 24 volt dc battery, using the NATO receptacle, as a back-up power source. Cables provided for this purpose, shall not be less than five meters long. The battery charger (see 3.1.2.1) shall be capable of charging a single GLPS battery, as well as simultaneously charging all batteries provided with the GLPS, from both 110 and 220 volt ac 50 and 60 hertz utility power sources, and also directly from a 24 volt dc military vehicle battery. The battery charger shall be capable of recharging the batteries with minimal degradation to their operational life span.

3.4.7 Illuminating target rod. The illuminating target rod shall be centered on the vertical axis of the optics and capable of being seen 360° through a howitzer's panoramic telescope at distances up to 400 meters during

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darkness, under clear visibility. The illumination shall be a red light adjustable from off to high intensity.

3.5 Environmental requirements

3.5.1 High and low temperatures. The GLPS shall perform as specified herein and show no indication of damage at ambient temperatures from +49° C to -46° C, when tested as specified in 4.6.3 and 4.6.4. The GLPS shall not exhibit any degradation of materials or performance after storage in its transport cases at ambient temperatures from +49° C to -46° C, when tested as specified in 4.6.3 and 4.6.4. Additionally, the presence of broken, cracked, loosened, or shifted parts or components; or opened joints; or moisture within any sealed component; or broken, splintered or deformed transport cases shall constitute failure of the test.

3.5.2 Wind. The GLPS shall perform as specified herein, unsheltered, in the presence of winds to 32.2 ±3.2 km/hr (20 ±2 mph), with wind gusts to 40.2 ±3.2 km/hr (25 ±2 mph), when tested as specified in 4.6.5.

3.5.3 Rain. The GLPS shall perform as specified herein and show no indication of damage or moisture penetration when tested as specified in 4.6.6.c, except that visibility through the optical devices and range determination with the laser rangefinder may be degraded. The transport case(s) shall be capable of protecting the items stored therein and show no indication of damage or moisture penetration when tested as specified in 4.6.6.b.

3.5.4 Icing/freezing rain. The GLPS shall perform as specified herein and show no indication of damage or moisture penetration when tested as specified in 4.6.7.b, except that visibility through the optical devices and range determination with the laser rangefinder may be degraded. The transport case(s) shall be capable of protecting the items stored therein and show no indication of damage or moisture penetration when tested as specified in 4.6.7.a.

3.5.5 Humidity. The GLPS shall perform as specified herein and show no indication of damage or moisture penetration when tested as specified in 4.6.8.b. The transport case(s) shall be capable of protecting the items stored therein and show no indication of damage or moisture penetration when tested as specified in 4.6.8.a.

3.5.6 Sand and dust. The GLPS shall perform as specified herein and show no indication of damage or sand or dust penetration when tested as specified in 4.6.9.1.b and 4.6.9.2.b, except that visibility through the optical devices and range determination with the laser rangefinder may be degraded. The transport case(s) shall be capable of protecting the items stored therein and show no indication of damage or sand or dust penetration when tested as specified in 4.6.9.1.a and 4.6.9.2.a.

3.5.7 Salt fog. The GLPS shall perform as specified herein and show no indication of corrosion, electrical, or physical effects with the potential to degrade its proper functioning after undergoing the salt fog test specified in 4.6.10.c. The transport case(s) shall be capable of protecting the items stored therein and show no indication of damage or moisture penetration when tested as specified in 4.6.10.b.

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3.5.8 Fungus. While in its transport case(s) and when tested as specified in 4.6.11, the GLPS, including all components, will not support fungus growth to any degree that would preclude its proper functioning.

3.6 Transportability requirements

3.6.1 Vibration. The GLPS shall perform as specified herein and show no indication of damage after experiencing vibrational levels normally encountered during transport in its transport case(s) via the High Mobility Multipurpose Wheeled Vehicle (HMMWV) stowed as secured-cargo and also as loose-cargo, as simulated by the vibration test specified in 4.7.1. The transport case(s) shall not be rendered unusable after the vibration test by showing evidence of delamination or ruptured seals, cracked or broken welds which impair structural integrity, or permit water leakage, or cause dysfunction of the hinges or latches, carrying devices, pressure relief valves, or mounting provisions inside the case(s).

3.6.2 Shock. The GLPS shall perform as specified herein and show no indication of damage after undergoing shock normally induced by loading and unloading the equipment while in its transport case(s), as simulated by the shock test specified in 4.7.2. The transport case(s) shall not be rendered unusable after the shock test by showing evidence of delamination or ruptured seals, cracked or broken welds which impair structural integrity, or permit water leakage, or cause dysfunction of the hinges or latches, carrying devices, pressure relief valves, or mounting provisions inside the case(s).

3.6.3 Low pressure (altitude). The GLPS shall be capable of being transported by air in its transport case(s) in a low-pressure environment and shall survive rapid decompression during air transport, when tested as specified in 4.7.3.

3.6.4 Airdrop. The GLPS shall perform as specified herein and show no indication of damage after being airdropped in its transport case(s) in a HMMWV, when tested as specified in 4.7.4.

3.7 Electromagnetic environment. The GLPS shall not be susceptible to electromagnetic interference from high tension electrical wires or other military equipment used in the same area; or radiate emissions that interfere with the operation of other military equipment used in the same area (see 6.2) when tested as specified in 4.8.

3.8 Safety and health. The laser rangefinder shall be a Class 1 laser in accordance with the latest version of ANSI Z136.1. The GLPS shall include safety features which eliminate or control potential hazards to personnel or equipment without impairing equipment performance. The GLPS shall not present hazards: with even a remote probability of causing death or system loss; with even occasional probability of causing severe injury or major damage to equipment; or with frequent probability of minor injury or minor damage to equipment. Other identified hazards shall be controlled to the lowest risk level possible. When required (see 6.2), a Safety Assessment Report (SAR) shall be provided. Safety and health shall be tested as specified in 4.9.

3.9 Human factors engineering. The GLPS design shall conform with Army accepted human factors engineering criteria (see 6.5) when tested as specified in 4.10. Emphasis shall be given, but not limited to minimizing the size and

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weight, control and display integration, visual displays, minimizing controls, labeling, anthropometry, design for maintainer, and safety and health.

a. The GLPS shall be designed for assembly and disassembly in the operational environments defined herein (see 3.5). Alignment guides or aids shall be provided on components, where feasible, to assist in assembly.

b. The GLPS shall be operable and maintainable in daylight and darkness by soldiers dressed in arctic clothing and MOPP IV.

c. The GLPS design shall permit the performance of the mission to required accuracies (see 3.4.2), and within required times (see 3.4.3 and 3.4.4).

d. The weight of any individual GLPS component, except for transport cases, shall not exceed 25 lbs. The weight of any individual transport case shall not exceed the maximum two person lift of 74 lbs when loaded. Transport cases exceeding 56 lbs shall be labeled "TWO-PERSON-LIFT". Transport case handles appropriate to a one or two person lift shall be provided.

e. The GLPS shall be capable of being carried, when removed from the transport case(s), by one person making no more than two trips for a distance up to 100 meters. The maximum one person lift and carry weight is 50 lbs.

f. The GLPS data control and display system shall be simple to operate and user friendly with menu driven prompts from the microprocessor for each function. The number of control and display function keys shall be minimized, clearly marked to identify their function, and located so that they may be operated and read without requiring the operator to assume an uncomfortable, awkward or unsafe position. The display shall be readily visible in all expected lighting conditions, including bright sunlight and new moon levels of darkness. The data control and display panel shall be equipped with a lighting system with adjustable illumination levels. Under nighttime operating conditions, there shall be no visible light emitting from the control and display system beyond 50 meters.

g. Repeated operator errors of omission or commission shall not exceed 5% during the mission performance test (see 4.5.1) for setup times and accuracies.

3.10 Identification and marking. System names and model numbers shall be restricted to GLPS identification plates. Plates shall be attached in a conspicuous protected location. Controls shall be clearly marked to identify their function. The laser rangefinder shall be marked in accordance with ANSI Z136.1. Transport cases shall be permanently marked to indicate they contain delicate instruments and to show proper orientation for transport.

3.11 Color. The color of the GLPS, including the components, shall be standardized with other miliary equipment with which it is used (see 6.6).

3.12 Reliability. The GLPS shall demonstrate a mean-time-between-system-abort (MTBSA) of 540 hours at a 80 percent lower confidence level, when tested as specified in 4.11.

3.13 Interchangeability. All components of the GLPS (see 3.1) having the same part number shall be functionally and dimensionally interchangeable

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between systems without alteration of the components themselves or of any other component or part, when tested as specified in 4.12.

3.14 Workmanship. The GLPS shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect its performance, life, serviceability, interchangeability, or appearance.

3.15 Government-loaned property. When specified (see 6.7), a PLGR will be loaned by the Government to the contractor.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3)

4.2 First article inspection. First article inspection shall be performed on GLPS samples when first article samples are required (see 6.2 and 6.3). This inspection shall include the examination in accordance with Table 1 and the tests marked "x" under "first article test" in Table 2.

4.3 Conformance inspection. A conformance inspection shall be performed on 100% of the completed production GLPS. Conformance inspection shall include the examination in accordance with Table 1 and the tests marked "x" under "conformance test" in Table 2.

4.4 Examination. Each GLPS shall be examined in accordance with Table 1. This element of inspection shall primarily encompass visual examinations and physical measurements. The capability of the transport case(s) to be decontaminated (see 3.1.1.6) shall be determined by analysis. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

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TABLE 1. Examination

No.	Defect	Requirement Paragraph
101.	Assembled GLPS not as specified	3.1
102.	Azimuth gyroscope not as specified.	3.1.1.1
103.	Electronic digital optical instrument not as specified.	3.1.1.2
104.	Laser Rangefinder not as specified.	3.1.1.3
105.	PLGR interface not as specified.	3.1.1.4
106.	Tripod not as specified.	3.1.1.5
107.	Transport case(s) not as specified.	3.1.1.6
108.	Battery/battery pack not as specified.	3.1.1.7
109.	Illuminating target rod not as specified.	3.1.1.8
110.	Battery charger not as specified.	3.1.2.1
111.	Power conditioner (when supplied) not as specified.	3.1.2.2
112.	Protective cover not as specified.	3.1.2.3
113.	Winterization kit (when provided) not as specified.	3.1.2.4
114.	Electric cables, receptacles, and fuses not as specified.	3.1.2.5
115.	Tripod base (when provided) not as specified.	3.1.2.6
116.	Tools, materials, supplies, and spares not as specified.	3.1.2.7
117.	Materials not as specified	3.3
118.	Material deterioration prevention and control not as specified	3.3.1
119.	Dissimilar metals are not effectively insulated from each other	3.3.2
120.	Contractor does not have documentation available for identification of materials, material finishes, or treatment	3.3.3
121.	Safety and health not as specified	3.8
122.	Human factors engineering not as specified	3.9
123.	Identification and marking not as specified	3.10
124.	Color not as specified.	3.11
125.	Interchangeability not as specified.	3.13
126.	Workmanship not as specified	3.14

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TABLE 2. Tests

First article test	Conformance test	Test	Test paragraph	Requirement paragraph
x		Mission performance	4.5.1	3.4 3.4.1 3.4.2
	x	Conformance acceptance	4.5.2	3.4 3.4.1 3.4.2
x		Extreme latitude accuracies	4.5.3	3.4.2.3
x		Setup time	4.5.4	3.4.3
x		Mission times	4.5.5	3.4.4
x		Microprocessing, data control and display	4.5.6	3.4.5
x		Battery/battery pack capacity	4.5.7	3.4.6
x		Back-up power	4.5.8	3.4.6
x		Illuminating target rod	4.5.9	3.4.7
x		High and low temperatures	4.6.3 4.6.4	3.5.1
x		Wind	4.6.5	3.5.2
x		Rain	4.6.6	3.5.3
x		Icing/freezing rain	4.6.7	3.5.4
x		Humidity	4.6.8	3.5.5
x		Sand and dust	4.6.9	3.5.6
x		Salt fog	4.6.10	3.5.7
x		Fungus	4.6.11	3.5.8
x		Vibration	4.7.1	3.6.1
x		Shock	4.7.2	3.6.2

(TABLE 2 continued next page)

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First article test	Confor-mance test	Test	Test para-graph	Require-ment paragraph
x		Low pressure (altitude)	4.7.3	3.6.3
x		Airdrop	4.7.4	3.6.4
x		Electromagnetic environment	4.8	3.7
x		Safety and health	4.9	3.8
x		Human factors engineering	4.10	3.9
x		Reliability	4.11	3.12
x		Interchangeability	4.12	3.13

4.5 Methods of Inspection

4.5.1 Mission performance

a. The mission performance test for GLPS first articles (see 3.2) shall be conducted during day and night conditions, and during inclement weather (rain or snow), if possible. Tests shall be conducted by personnel wearing the combat uniform and MOPP 0 and MOPP IV. A 1 x 90 meter network of 10 third order survey control points shall be established and used as orienting stations. Six howitzers or simulated howitzer targeting devices shall be used. The GLPS shall be set up for operation at the orienting stations. The operator shall sight to the pantel lenses of the six howitzers that are located 30 to 400 meters away from the orienting stations. The grid azimuth/ deflection, slope distance, vertical angle, coordinates, altitudes, and distances for each howitzer as displayed by the GLPS will be recorded. The time required to lay the first and subsequent five howitzers shall be recorded repeatedly until the required sample size has been collected. The start time for the first howitzer shall be when power is applied to the gyroscope and the stop time shall be when the deflection and location of the first howitzer are displayed. The laying of six (6) howitzers from one orienting station represents one mission (see 6.4.3).

b. Using the back polar plot method, the GLPS shall determine the position of orienting stations from visible survey control points located at ranges 30 to 2000 meters from the GLPS.

c. The GLPS shall be operated until it is determined that a MTBSA of 540 hours (see 4.11) at an 80 percent lower confidence level is achieved.

d. Nonconformance with 3.4 or 3.12 shall constitute failure of this test.

4.5.2 Conformance acceptance. A conformance acceptance test shall be performed for each GLPS production item during daylight in good weather conditions. The conformance acceptance test shall consist of the following:

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a. One (1) third order survey control point shall be established and used as an orienting station.

b. One (1) third order survey control point shall be established and used as a target. The target, approximately one foot square in size, shall be positioned approximately 100 meters from the orienting station.

c. The GLPS shall be removed from its transport case(s) and set up for operation at the orienting station.

d. The PLGR shall be connected to the GLPS and the accuracy of data transmission verified.

e. Using positional input of the orienting station established in 4.5.2.a, and an azimuth of fire approved by the Government, the operator shall sight to the target. The grid azimuth/deflection, slope distance, vertical angle, coordinates, altitude, and distance for the target as displayed by the GLPS shall be recorded.

f. The time to derive the position and deflection to the target shall be recorded.

g. Nonconformance with 3.4.1, 3.4.2, or 3.4.4 shall constitute failure of this test.

4.5.3 Extreme latitude accuracies. A high latitude accuracy test shall be conducted at a test facility to be specified (see 6.2) between 65° and 75° North latitude under the existing ambient temperature at the time of test. The GLPS shall be positioned over an orienting station for which a third order azimuth is established. Not less than 10 back polar plots and azimuth determinations shall be performed. Nonconformance with 3.4.2.1 and 3.4.2.3 shall constitute failure of this test.

4.5.4 Setup time. During the mission performance test (see 4.5.1), the time to set up the GLPS for operation shall be recorded repeatedly until the required sample size has been collected (setup is defined in 6.4.2). Non-conformance with 3.4.3 shall constitute failure of this test.

4.5.5 Mission times. During the mission performance test (see 4.5.1), the time to derive position and deflection to the first howitzer and then each subsequent howitzer shall be recorded repeatedly until the required sample size has been collected. Failure to meet times specified in 3.4.4 shall constitute failure of this test.

4.5.6 Microprocessing, data control and display. During the mission performance test (see 4.5.1), the GLPS microprocessing and data control and display functions shall be inspected. Nonconformance with 3.4.5 shall constitute failure of this test.

4.5.7 Battery/battery pack capacity. During the mission performance test (see 4.5.1), the GLPS shall be operated for at least 20 missions without changing or recharging the battery/battery pack. The test shall begin with a fully charged battery/battery pack and shall be repeated (with a fully charged battery/battery pack each time) until the required sample size has been collected. Nonconformance with 3.4.6 shall constitute failure of this test.

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4.5.8 Back-up power. During the mission performance test (see 4.5.1), the GLPS shall be operated using power supplied directly from a military vehicle 24 volt dc battery. The battery charger shall be operated to charge the GLPS batteries from both 110 and 220 volt ac 50 and 60 hertz utility power sources, and also from a 24 volt dc military vehicle battery. Nonconformance with 3.4.6 shall constitute failure of this test.

4.5.9 Illuminating target rod. During the mission performance test (see 4.5.1), the GLPS shall be operated during darkness, under clear visibility. Nonconformance with to 3.4.7 shall constitute failure of this test.

4.6 Environmental

4.6.1 Performance checks before and after tests. Performance checks of the GLPS shall be conducted before and after each test specified herein. Performance checks shall be conducted during daylight in good weather conditions and include the following:

a. One (1) third order survey control point shall be established and used as an orienting station.

b. One (1) third order survey control point shall be established and used as a target. The target, approximately one foot square in size, shall be positioned approximately 100 meters from the orienting station.

c. The GLPS shall be removed from its transport case(s) and set up for operation at the orienting station.

d. The PLGR shall be connected to the GLPS and the accuracy of data transmission verified.

e. Using positional input of the orienting station established in 4.6.1.a, and upon entering an azimuth of fire, the operator shall sight to the target. The grid azimuth/deflection, slope distance, vertical angle, coordinates, altitude, and distance for the target as displayed by the GLPS shall be recorded.

f. The time to set up and derive the position and deflection to the target shall be recorded.

g. Nonconformance with 3.4.1, 3.4.2, 3.4.3, or 3.4.4 after a test shall constitute failure of that test.

4.6.2 Performance checks during exposure. Limited access to test items may restrict performance checks during exposure to environments while undergoing environmental tests; but, to the extent feasible, the GLPS shall be set up and operated as specified in 4.6.1 to verify system performance. As a minimum, the following performance checks shall be performed during exposure to environments:

a. Verifying that the GLPS can be assembled and disassembled.

b. Verifying that all electrical systems are functioning.

c. Verifying that the microprocessor is functioning.

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d. Verifying that the data control and display system is functioning.

The protective cover (see 3.1.2.3) shall not be used to protect the GLPS when exposed to environments during environmental tests.

4.6.3 High temperature. Performance checks before the high temperature test shall be performed as specified in 4.6.1.

a. For the storage test, the GLPS, including all components, shall be stored in its transport case(s) in the climatic chamber for a total of 168 hours (seven cycles of 24 hour periods). Temperature cycles are described in Table 3. Following exposure, the GLPS shall be removed from its case(s) and set up for operation. Performance checks shall be performed as specified in 4.6.1. Nonconformance with 3.5.1 shall constitute failure of this test.

Table 3. High Temperature Cycle

Hour of test	Temp		Humidity %	Hour of test	Temp		Humidity %
	°C	°F			°C	°F	
01	35	95	6	13	47	116	3
02	34	94	7	14	48	118	3
03	34	93	7	15	48	118	3
04	33	92	8	16	49	120	3
05	33	91	8	17	48	118	3
06	32	90	8	18	48	118	3
07	33	91	8	19	46	114	3
08	35	95	6	20	42	108	4
09	38	101	6	21	41	105	5
10	41	106	5	22	39	102	6
11	43	110	4	23	38	100	6
12	44	112	4	24	37	98	6

b. During the storage test, during three of the seven cycles at +49° C, the GLPS shall be removed from its case(s) and performance checks during exposure (see 4.6.2) shall be performed. Nonconformance with 3.5.1 shall constitute failure of this test.

4.6.4 Low temperature. Performance checks before the low temperature test shall be performed as specified in 4.6.1.

a. During the storage test, the GLPS, including all components, shall be stored in its transport case(s) in the climatic chamber for 24 hours (1

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cycle). The temperature cycle is described in Table 4. Following exposure during storage, the GLPS shall be removed from its case(s) and set up for operation. Performance checks shall be performed as specified in 4.6.1. Nonconformance with 3.5.1 shall constitute failure of this test.

Table 4. Low Temperature Cycle

Hour of test	Temp		Hour of test	Temp	
	°C	°F		°C	°F
01	-21	-6	13	-46	-51
02	-23	- 9	14	-46	-51
03	-25	-13	15	-46	-51
04	-27	-17	16	-44	-47
05	-30	-22	17	-42	-44
06	-32	-26	18	-40	-40
07	-35	-31	19	-37	-35
08	-37	-35	20	-34	-29
09	-40	-40	21	-31	-24
10	-42	-44	22	-27	-17
11	-44	-47	23	-24	-11
12	-46	-51	24	-22	-8

b. During the storage test, the GLPS shall be removed from its transport case(s) at -46° C and manipulated to determine whether it can be made operational without affecting its functional performance. Following manipulation, performance checks during exposure (see 4.6.2) shall be performed. Tests shall be conducted by personnel wearing arctic clothing. The use of a winterization kit, if provided, shall be as prescribed in the operators manual. Nonconformance with 3.5.1 shall constitute failure of this test.

4.6.5 Wind

a. Performance checks before and after the wind test shall be performed as specified in 4.6.1.

b. The GLPS shall be set up at normal operating height and subjected to a simulated wind directed horizontally at the instrument for the duration of one mission (see 6.4.3). Tripod feet shall be firmly planted into the ground. Wind speed shall be 32.2 ±3.2 km/hr (20 ±2 mph), with wind gusts to 40.2 ±3.2 km/hr (25 ±2 mph) at a rate of 5 to 15 gusts per minute. Performance checks during exposure shall be performed as specified in 4.6.2. Nonconformance to 3.5.2 shall constitute failure of this test.

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4.6.6 Rain. Performance checks before the rain test shall be performed as specified in 4.6.1.

a. The rain shall be produced by a water distribution device of such design that water is emitted in the form of droplets having a diameter range predominantly between 0.5 and 4.5 mm.

b. For the storage test, the rain shall be dispersed completely over the transport case(s) with the GLPS, including all components, stored therein. The rain shall be accompanied by wind with the rate of up to 64.4 km/hr (40 mph). Exposure time shall be one (1) hour per side of the transport case(s). Following exposure during storage, the GLPS shall be removed from its case(s) and set up for operation. Performance checks shall be performed as specified in 4.6.1. Nonconformance with 3.5.3 shall constitute failure of this test.

c. Performance checks during exposure shall be performed as specified in 4.6.2. During exposure, the GLPS shall be removed from its transport case(s), set up for operation, used to lay six targets, and put back in its transport case(s) within 16 minutes. This exposure cycle shall be repeated with the rain directed at each side of the GLPS. The rain shall be dispersed completely over the GLPS while accompanied by wind with the rate of up to 32.2 km/hr (20 ±2 mph). Nonconformance with 3.5.3 shall constitute failure of this test.

4.6.7 Icing/freezing rain. Performance checks before the icing/freezing rain test shall be performed as specified in 4.6.1.

a. For the storage test, the temperature of the GLPS, including all components, stored in its transport case(s) shall be stabilized at approximately 2 ±1° C. A uniform rain spray of prechilled water shall be dispersed completely over the transport case(s) for 1 hour. The chamber air temperature shall be lowered to -10° C (+14° F) and the rain rate maintained until 13 mm (1/2 in) of ice has accumulated on all sides and on the upper surface. With the rain stopped, the chamber air temperature shall be adjusted to -6° C (+21° F) for 2 to 6 hours and the GLPS shall be removed from its case(s). Ice removal is limited to integral methods plus simple, expedient, and obvious methods. Following exposure during storage, performance checks shall be performed as specified in 4.6.1. Nonconformance with 3.5.4 shall constitute failure of this test.

b. Performance checks during exposure shall be performed as specified in 4.6.2. During exposure, the GLPS shall be removed from its transport case(s), set up for operation, used to lay six targets, and put back in its transport case(s) within 16 minutes. During this test, the chamber air temperature shall be -10° C (+14° F) and pre-chilled water shall be dispersed completely over the instrument. Ice removal is limited to integral methods plus simple, expedient, and obvious methods. Nonconformance with 3.5.4 shall constitute failure of this test.

4.6.8 Humidity. Performance checks before the humidity test shall be performed as specified in 4.6.1.

a. For the storage test, the GLPS, including all components, shall be stored in its transport case(s) and placed in the climatic chamber. The GLPS shall be preconditioned at 23° C (73° F) with 50% humidity for 24 hours prior to starting the test. Following this, the chamber temperature shall be raised and the GLPS subjected to 15 cycles (360 hours) of high humidity as shown in

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Table 5. Following exposure during storage, the GLPS shall be removed from its case(s) and set up for operation. Performance checks shall be performed as specified in 4.6.1. Nonconformance with 3.5.5 shall constitute failure of this test.

Table 5. High Humidity Cycle

Hour of test	Temp		Humidity %	Hour of test	Temp		Humidity %
	°C	°F			°C	°F	
01	27	80	100	13	34	94	74
02	27	80	100	14	35	95	74
03	26	79	100	15	35	95	74
04	26	79	100	16	34	94	76
05	26	79	100	17	33	92	79
06	26	70	100	18	32	90	82
07	27	81	94	19	31	87	81
08	29	84	88	20	29	85	91
09	31	87	82	21	28	83	95
10	32	89	79	22	28	82	96
11	33	92	77	23	27	80	100
12	34	94	77	24	27	80	100

b. During the storage test, at least once every five cycles, the GLPS shall be removed from its case(s) and set up for operation. Performance checks during exposure (see 4.6.2) shall be performed. Nonconformance with 3.5.5 shall constitute failure of this test.

4.6.9 Sand and dust

4.6.9.1 Dust. Performance checks before the dust test shall be performed as specified in 4.6.1.

a. For the storage test, the GLPS, including all components, shall be stored in its transport case(s) and placed in the climatic chamber. The chamber temperature shall be set at 23° C (73° F) with less than 30% humidity, and the air velocity adjusted at 32.2 km/hr (20 mph). The dust feed control shall be adjusted for a dust concentration of 10,6 ±7 g/cu m (0.3 + 0.2 g/cu ft). These conditions shall be maintained for six hours, at which time the dust feeder shall be stopped, air velocity reduced, and the internal air temperature raised to 49° C (120° F). With the internal air temperature stabilized at 49° C, the dust feeder shall be started, with the same concentration as previously described above, and the test item shall be subjected to this phase of testing for an additional six hours. Following exposure during

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storage, the GLPS shall be removed from its case(s) and set up for operation. Performance checks shall be performed as specified in 4.6.1. Nonconformance with 3.5.6 shall constitute failure of this test.

b. The GLPS shall be removed from its transport case(s), set up for operation, and exposed to blowing dust for 16 minutes at ambient temperature. Following exposure, performance checks shall be performed as specified in 4.6.1. Dust removal is limited to integral methods plus simple, expedient, and obvious methods. Nonconformance with 3.5.6 shall constitute failure of this test.

4.6.9.2 Sand. Performance checks before the sand test shall be performed as specified in 4.6.1.

a. For the storage test, the GLPS, including all components, shall be stored in its transport case(s). The sand feeder shall be adjusted for a sand concentration which results in approximately 273 kg (600 lb) of sand being used in each 90-minute test duration, or an average of 3.03 kg/min (6.7 lb/min), at an average wind speed of 64.4 km/hr (40 mph). The test items shall be tested for 90 minutes each on all facing sides. Following exposure during storage, the GLPS shall be removed from its case(s) and set up for operation. Performance checks shall be performed as specified in 4.6.1. Nonconformance with 3.5.6 shall constitute failure of this test.

b. Performance checks during exposure shall be performed as specified in 4.6.2. During exposure, the GLPS shall be removed from its transport case(s), set up for operation, used to lay six targets, and put back in its transport case(s) within 16 minutes. This exposure cycle shall be repeated with the sand directed at each side of the GLPS at an average wind speed of 32.2 km/hr (20 mph). Nonconformance with 3.5.6 shall constitute failure of this test.

4.6.10 Salt fog. Performance checks before the salt fog test shall be performed as specified in 4.6.1.

a. The salt used for this test shall be sodium chloride containing (on a dry basis) not more than 0.1 percent sodium iodide and not more than 0.5 percent total impurities. Unless otherwise specified, a 5 ±1 percent solution shall be prepared by dissolving 5 parts by weight of salt in 95 parts by weight of water. The solution shall be adjusted to, and maintained at, the correct specific gravity by using the measured temperature and density of the salt solution. Sodium tetraborate (borax) may be added to the salt solution as a pH stabilization agent in a ratio not to exceed 0.7g sodium tetraborate to 75 liters of salt solution. The pH of the salt solution, as collected as fallout in the chamber, shall be maintained between 6.5 and 7.2 with the solution temperature at 35° C (95° F). Only diluted chemically pure hydrochloric acid or chemically pure sodium hydroxide shall be used to adjust pH.

b. For the storage test, the GLPS, including all components, shall be stored in its transport case(s) and placed in the test chamber. The chamber temperature shall be adjusted to 35° (95° F) to condition the GLPS for at least two hours before introducing the salt fog. The salt solution shall be continuously atomized into the chamber for a period of at least 48 hours. During the entire exposure period, the salt fog fallout rate and pH of the fallout solution shall be measured at least at 24 hour intervals. Fallout shall be between 0.5 and 3 ml/80cm²/hr. The GLPS shall then be stored in a standard ambient atmosphere for at least 48 hours for drying. Following

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exposure during storage, the GLPS shall be removed from its case(s) and set up for operation. Performance checks shall be performed as specified in 4.6.1. Nonconformance with 3.5.7 shall constitute failure of this test.

c. Performance checks during exposure shall be performed as specified in 4.6.2. During exposure, the GLPS shall be removed from its transport case(s), set up for operation, and exposed to the salt fog for 16 minutes. Nonconformance with 3.5.7 shall constitute failure of this test.

4.6.11 Fungus. Performance checks before the fungus test shall be performed as specified in 4.6.1.

a. The test fungi shall be Aspergillus niger, Aspergillus flavus, Aspergillus versicolor, Penicillium funiculosum, and Chaetopium globsum.

b. The GLPS, including all components, shall be placed in the chamber at 24° to 31° C (75° to 80° F) and 95 ±5% relative humidity for at least four hours immediately before inoculation. The test item and cotton fabric chamber control items shall be inoculated with the mixed fungal spore suspension by spraying the suspension on the control items and on and into the test item (if not permanently or hermetically sealed) in the form of a fine mist from an atomizer or nebulizer. The test items shall be replaced in their transport cases without tightening the fasteners (so that air can penetrate). Incubation shall start immediately following inoculation and the test duration shall be 28 days.

c. The test items shall be incubated under a daily cycle of temperature and humidity conditions consisting of 20 hours at a relative humidity of 95 ±5% and an air temperature of 30° ±1° C (85° ±2° F) followed by a 4-hour period in which condition of 95 ±5% relative humidity at 25° ±1° C (77° ±2° F) are maintained for at least two hours. Up to a total of two hours of the 4-hour period will be used for the transitions of temperature and relative humidity. Temperature and humidity conditions during the transition periods shall be as follows: temperature 24° to 31° C (75° to 88° F) and relative humidity above 90%. The 24-hour daily cycle shall be repeated for the test duration. After seven days, the growth on the control cotton strips shall be inspected to verify that the environmental conditions in the chamber are suitable for growth.

d. Following the 28-day incubation period, the GLPS shall be removed from its case(s) and set up for operation. Performance checks shall be performed as specified in 4.6.1. Nonconformance with 3.5.8 shall constitute failure of this test.

4.7 Transportability

4.7.1 Vibration. Performance checks before the vibration test shall be performed as specified in 4.6.1.

a. The GLPS, including all components, when in its transport case(s), shall be mounted on the vibration table and subjected to the vibration schedule in Table 6. The secured cargo portion of the test will be a simulated distance of 500 miles. The loose cargo portion will simulate 150 miles.

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Table 6. Simulated Road Vibration Course Profile

SURFACE/COURSE	LENGTH		SPEED	
	m	ft	km/hr	mph
Belgian Block	1202	3940	32	20
Radial Washboard	74	243	24	15
3-Inch Spaced Bump	233	764	32	20

b. Following exposure to each vibration test, the GLPS shall be removed from its case(s) and set up for operation. Performance checks shall be performed as specified in 4.6.1. Nonconformance with 3.6.1 shall constitute failure of this test.

4.7.2 Shock. Performance checks before the shock test shall be performed as specified in 4.6.1.

a. The GLPS, including all components, when in its transport case(s), shall be dropped onto a grassy ground. The height of drop can be found in Table 7.

Table 7. Transit Drop Test

Weight of Test Item and Case	Largest Dimensions	Height of Drop
Under 45.4 Kg (100 lbs)	Under 91 cm (35.8 in)	122 cm (48 in)
	91 cm and over	76 cm (30 in)

b. The cases shall be dropped on each face, edge, and corner. The total number of drops per case shall be 26. Drops shall be made from a quick-release hook, or drop tester. The cases shall be so oriented that upon impact, a line from the struck corner or edge to the center of gravity of the case and contents is perpendicular to the impact surface.

c. Following the 26 drops, the GLPS shall be removed from its case(s) and set up for operation. Performance checks shall be performed as specified in 4.6.1. Nonconformance with 3.6.2 shall constitute failure of this test.

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4.7.3 Low pressure (altitude). Performance checks before the low pressure test shall be performed as specified in 4.6.1.

a. For the storage test, the GLPS, including all components, shall be stored in its transport case(s) and placed in the test chamber. The chamber air pressure shall be adjusted at one or more rates specified for minimum cargo compartment pressures for military transport aircraft, ranging from 8.29 psia (57.2 kPa) to 10.71 psia (73.8 kPa) with equivalent altitudes ranging from 8,000 ft (2,400m) to 15,000 ft (4,570 m). Each air pressure tested shall be maintained for a minimum of one hour. Following the exposure to each low pressure tested, the GLPS shall be removed from its case(s) and set up for operation. Performance checks shall be performed as specified in 4.6.1. Nonconformance with 3.6.3 shall constitute failure of this test.

b. For the rapid decompression test, the GLPS, including all components, shall be stored in its transport case(s) and placed in the test chamber. The chamber air pressure shall be reduced at the rate specified to the maximum altitude of the anticipated aircraft. The pressure shall be reduced to an equivalent altitude of 12,200 m (40,000 ft) (18.8 kPa) as quickly as possible but in not more than 15 seconds, and this reduced pressure maintained for at least 10 minutes. Following the rapid decompression, the GLPS shall be removed from its case(s) and set up for operation. Performance checks shall be performed as specified in 4.6.1. Nonconformance with 3.6.3 shall constitute failure of this test.

4.7.4 Airdrop. Performance checks before and after each airdrop test shall be performed as specified in 4.6.1.

a. The GLPS shall be subject to one or more Simulated Airdrop Impact Tests. The GLPS, including all components, in its transport case(s) in a HMMWV rigged with an energy dissipator kit shall be free dropped from a 12.6 feet height to replicate 28.5 fps impact.

b. The GLPS shall be subject to three actual low velocity air drops. The GLPS, including all components, in its transport case(s) in a HMMWV rigged with an energy dissipator kit shall be air dropped from a military aircraft using one of the current family of cargo parachutes. A typical low velocity airdrop is no more than 28.5 fps.

c. Nonconformance with 3.6.4 shall constitute failure of this test.

4.8 Electromagnetic environment. The GLPS shall be subjected to electromagnetic interference and susceptibility tests to determine if it radiates emissions that interfere with the operation of other military equipment used in the same area; or, is susceptible to electromagnetic interference from high tension electrical wires or other military equipment used in the same area. The military equipment used in the same area as the GLPS shall be identified when required (see 6.2). Nonconformance with 3.7 shall constitute failure of this test.

4.9 Safety and health. When required (see 6.2), a Safety Assessment Report (SAR) shall be provided by the contractor. The SAR shall be reviewed by the Government prior to first article inspection (see 4.2). A safety checklist shall be prepared by the Government and used for inspection of the GLPS. Nonconformance with 3.8 shall constitute failure of this test.

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4.10 Human factors engineering. Human factors characteristics shall be evaluated during the first article inspection (see 4.2). Nonconformance with 3.9 shall constitute failure of this test.

4.11 Reliability. The GLPS shall be operated for 869 hours, and if there are no system aborts, the reliability requirement in 3.12 will have been met. If there are one or more system aborts during the 869 hours, then GLPS operations shall be continued until a number of hours needed are collected that correspond to the allowable number of system aborts for that number of hours; up to 4,271 hours at which time five or less system aborts are allowed for the reliability requirement to be met. Nonconformance with 3.12 shall constitute failure of this test.

4.12 Interchangeability. Performance checks before the interchangeability test shall be performed as specified in 4.6.1.

a. An interchangeability matrix shall be developed for major components of the first article GLPSs. Major components shall be interchanged between GLPS in accordance with the matrix and operated to verify fit and performance of each configuration.

b. Nonconformance to 3.13 shall constitute failure of this test.

5. PACKAGING

5.1 Packaging requirements. Packaging requirements shall be as specified in the contract or order (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The GLPS is intended for use in establishing a reference line from which azimuths to various points may be quickly determined in artillery fire-control. The GLPS will give firing batteries all weather autonomous positioning and directional capability by rapidly determining azimuth, deflection, and position (coordinates and altitude) of each howitzer from one centrally located orienting station.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1, 2.2, and 2.3).

c. When first articles are required for inspection and the number of first articles required (see 3.2).

d. When Government-loaned property is required (see 3.15).

e. Level of preservation and packing required (see 5.1).

f. When a tripod base is required and the number required (see 3.1.2.6).

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g. When a winterization kit is required (if proposed as a component of the GLPS) and the number required (see 3.1.2.4).

h. When inspections are to be performed by the Government.

i. When inspections are to be performed by the contractor.

j. When Government facilities are to be used for inspections performed by the contractor.

k. The test facilities to be used for extreme latitude accuracy tests (see 4.5.3).

l. When the contractor shall propose menu driven prompts and visual displays to the Government for approval (see 3.4.5).

m. When the contractor shall identify the specific material, material finish or treatment for use with component and sub-components and make such information available to the Government (see 3.3.3).

n. What military equipment is used in the same area as the GLPS (see 3.7).

o. When a Safety Assessment Report (SAR) is required (see 3.8).

p. When data are required.

6.3 First Article. When first articles are required, the items shall be initial production samples and include all major and additional components (see 3.1.1 and 3.1.2). Acquisition documents should include specific instructions regarding arrangements for examination and tests, approval procedures, and disposition of the first articles after tests.

6.4 Definitions

6.4.1 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials.

6.4.2 Setup. Setup is defined as the removal of components from the case(s) and assembly and leveling of the GLPS, after reaching the orienting station.

6.4.3 Mission. One mission equals the GLPS set-up and deriving position and deflection for a battery of six (6) howitzers.

6.5 Human factors engineering. MIL-STD-1472 can be used as a reference for Army accepted human factors engineering criteria.

6.6 Color. The color of military equipment used with GLPS is olive drab No. 24064 as described in FED-STD-595.

6.7 Government-loaned property. The contracting officer will arrange to loan the property listed in 3.15 when applicable (see 6.2).

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6.8 Provisioning. The contracting officer will include provisioning requirements for repair parts and maintenance tools as necessary (including special tools), and instructions on shipment of GLPS.

6.9 Subject term (key word) listing

Gyroscope
Laser Rangefinder
PLGR
Optics
Howitzer
Survey
Deflection

Custodian:
Army - CE
Navy - MC

Review activity:
Navy - AS
DLA - GS

Preparing activity:
Army - CE

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